

**Dept. of Electrical & Computer Engineering (ECE),
Rajshahi University of Engineering & Technology**

Course Code: ECE 2216

Course Title: Data Base Systems Sessional

Instructed by:

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Lecturer, ECE, RUET

Course Outcomes (COs):

COs	Description	POs
1	Execute the basic queries of a database language.	PO5
2	Develop a real-life database application.	PO9

Marks Distribution:

Assessment Tools		Marks (%)	
Continuous Assessment (CA)	Lab Participation	10%	40%
	Continuous Assessment	10%	
	Project	10%	
	Lab Report	10%	
Summative Assessment (SA)	Quiz	20%	60%
	Lab Final	15%	
	Board Viva	25%	
Total		100%	

Lab – 1:

Experiment Name: Creating and Managing Databases using Data Definition Language (DDL) and Data Manipulation Language (DML).

Data Definition Language (DDL): DDL is the short name for Data Definition Language, which deals with database schemas and descriptions of how the data should reside in the database.

Data Manipulation Language (DML): DML focuses on manipulating the data stored in the database, enabling users to retrieve, add, update and delete data.

Data Query Language (DQL)

DQL is a subset of SQL used to retrieve data from a database without modifying it. Its main command is `SELECT`, which allows users to fetch specific information based on their requirements.

- **CREATE:** To create a database and its objects like (table, index, views, stored procedure, function and triggers). - DDL
 - ✓ **CREATE DATABASE:** CREATE DATABASE database_name;
 - ✓ **CREATE TABLE:** CREATE TABLE table_name(
column1_name data_type constraint,
column2_name data_type constraint,
column3_name data_type);
- **INSERT:** insert data into a table - DML
 - ✓ **INSERT INTO:** INSERT INTO table_name (column1, column2, column3)
VALUES (value1, value2, value3);
- **SELECT:** to fetch specific information based on their requirements. -DQL
 - ✓ **SELECT:** SELECT * FROM table name

- **ALTER:** alters the structure of the existing database - DDL
- **RENAME:** rename an object - DDL
 - ✓ **RENAME TABLE:** ALTER TABLE old_table_name RENAME TO new_table_name;
 - ✓ **RENAME COLUMN:** ALTER TABLE table_name RENAME COLUMN old_name TO new_name;
 - ✓ **CHANGE:** ALTER TABLE table_name CHANGE old_column_name new_column_name DATA_TYPE;
 - ✓ **MODIFY:** ALTER TABLE table_name MODIFY COLUMN column_name NEW_DATA_TYPE;
 - ✓ **ADD:** ALTER TABLE table_name
ADD column_name data_type;
- **COMMENT:** add comments to the data dictionary - DDL
 - /* this is a comment*/
- **DELETE:** Delete all records from a database table - DML
 - ✓ **DELETE:** DELETE FROM table_name WHERE condition;
- **DROP:** delete objects from the database - DDL
 - ✓ **DROP COLUMN:** ALTER TABLE table_name DROP COLUMN column_name;
 - ✓ **DROP TABLE:** DROP TABLE IF EXISTS users;
- **TRUNCATE:** remove all records from a table, including all spaces allocated for the records. - DDL
 - TRUNCATE:** TRUNCATE TABLE table_name;

Lab Work:

Student_Info

Student_ID	Name	Gender	Dept_Name	CGPA	Email
101	Rahim	Male	CSE	3.85	rahim@gmail.com
102	Karim	Male	EEE	3.65	karim@gmail.com
103	Nadia	Female	BBA	3.92	nadia@gmail.com
104	Rina	Female	CSE	3.55	rina@gmail.com
105	Sohan	Male	ME	3.70	sohan@gmail.com

1. Create a table named **Student_Info**.
2. Insert all the above records into the **Student_Info** table.
3. Display all records from the Student_Info table.
4. Display only the **Student_ID**, **Name**, and **CGPA** columns.
5. Display the information of students from the **CSE** department.
6. Change the department of student **Sohan** from **ME** to **EEE**.
7. Delete the record of the student whose **Student_ID = 104/your ID**.
8. Display the final contents of the **Student_Info** table.